

DIABETES PREDICTION USING MACHINE LEARNING

Data Science and Machine Learning Project Report

Student Name	Bharathi M
Project Title	Diabetes Prediction
Domain	Data Science and Machine Learning
Programming Language	Python
Machine Learning Model	Logistic Regression
Repository	upskillcampus

1. Abstract

Diabetes Prediction is a machine learning project developed to predict whether a person is likely to have diabetes based on medical and demographic measurements. The project uses the Pima Indians Diabetes Dataset and Logistic Regression. Data preprocessing, feature standardization, model training and evaluation are performed using Python and scikit-learn.

2. Introduction

Diabetes is a common health condition in which blood glucose levels become too high. Machine learning can be used to identify patterns in patient data and support early risk prediction. This project demonstrates a simple classification workflow from data loading to prediction.

3. Objectives

- To understand and preprocess a diabetes dataset.
- To divide the data into training and testing sets.
- To train a Logistic Regression classification model.
- To evaluate the model using accuracy, classification report and confusion matrix.
- To predict diabetes for a new sample using the trained model.

4. Dataset Description

Feature	Description
Pregnancies	Number of pregnancies
Glucose	Plasma glucose concentration
BloodPressure	Diastolic blood pressure
SkinThickness	Triceps skin fold thickness
Insulin	2-Hour serum insulin
BMI	Body Mass Index
DiabetesPedigreeFunction	Diabetes family-history score
Age	Age of the person
Outcome	0 = No diabetes, 1 = Diabetes

5. Technologies Used

- Python – programming language
- Pandas – data loading and manipulation
- NumPy – numerical operations
- Scikit-learn – preprocessing, model building and evaluation
- Jupyter Notebook / VS Code – development environment
- GitHub – project code and report repository

6. Methodology

1. Load the diabetes dataset.
2. Inspect the records, shape and missing values.
3. Replace invalid zero values in selected medical measurements with missing values and fill them using the median.
4. Separate input features (X) and target variable (y).
5. Split the dataset into 80% training data and 20% testing data.
6. Standardize numerical features using StandardScaler.
7. Train a Logistic Regression model.
8. Predict the test data and calculate evaluation metrics.
9. Use the trained model to predict a new sample.

7. Machine Learning Model

Logistic Regression is a supervised classification algorithm. It estimates the probability of a binary outcome and is suitable for predicting the two classes in this project: diabetes present (1) and diabetes not present (0).

8. Data Preprocessing

Some medical attributes contain zero values that do not represent realistic measurements. These values are treated as missing and replaced with the median of the corresponding feature. The features are then standardized so that variables with different numerical scales can be used effectively by the Logistic Regression model.

9. Training and Testing

The dataset is divided using an 80:20 ratio. The training set is used to learn the model, while the test set is kept separate for evaluation. Stratified splitting maintains a similar class distribution in both sets.

10. Evaluation

The program displays accuracy, a classification report and a confusion matrix. The exact values can vary slightly depending on the dataset version and preprocessing. The report therefore avoids claiming a fixed accuracy before the code is executed.

11. Sample Prediction

The program contains an example patient record. After scaling the sample using the same scaler fitted on the training data, the model predicts whether diabetes is likely and prints the estimated probability.

12. Advantages

- Simple and easy-to-understand machine learning workflow.
- Uses standard data preprocessing techniques.

- Provides measurable model evaluation.
- Can be extended with other classification algorithms.

13. Limitations

- Prediction quality depends on the dataset and its features.
- The model is not a substitute for professional medical diagnosis.
- The project uses a relatively small benchmark dataset.

14. Future Scope

- Compare Logistic Regression with Random Forest, SVM, XGBoost and other models.
- Perform hyperparameter tuning and cross-validation.
- Build a user-friendly web application for prediction.
- Use a larger and more diverse clinical dataset.

15. GitHub Repository Details

Repository name: **upskillcampus**

Code file: **DiabetesPrediction.py**

Report file: **DiabetesPrediction_Bharathi_USC_UCT.pdf**

Replace the placeholders below with your actual GitHub username before uploading the final report.

Code link:

<https://github.com/YOUR-USERNAME/upskillcampus/blob/main/DiabetesPrediction.py>

Report link:

https://github.com/YOUR-USERNAME/upskillcampus/blob/main/DiabetesPrediction_Bharathi_USC_UCT.pdf

16. Conclusion

The Diabetes Prediction project demonstrates how machine learning can be applied to a binary classification problem. The workflow covers data preprocessing, feature scaling, Logistic Regression, model evaluation and prediction. The project can be further improved by testing additional models and using larger datasets.